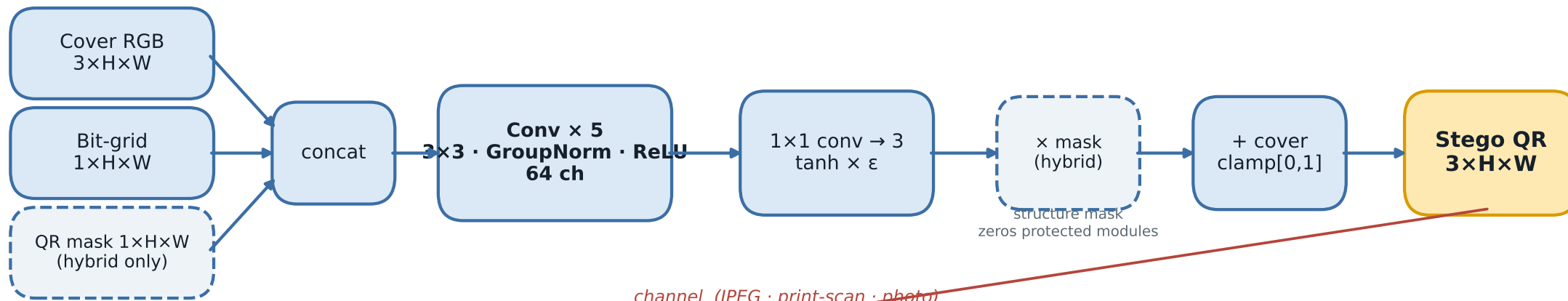
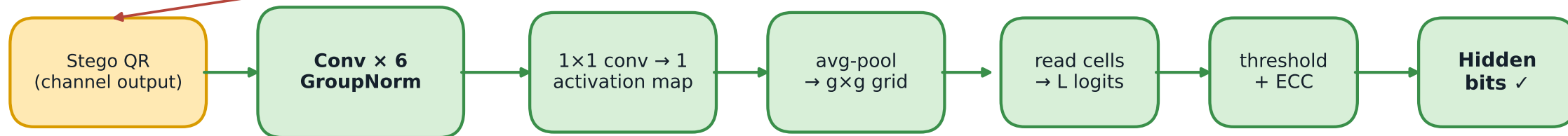


Encoder



Decoder



Spatial bit-grid: each of the L payload bits occupies one cell of a $\lceil\sqrt{L}\rceil \times \lceil\sqrt{L}\rceil$ grid upsampled to H×W; the decoder average-pools back to the grid and reads each cell.

Modes — segregated: 3 independent per-channel sub-networks · cross-channel: joint RGB (shown) · hybrid: joint + structural mask + bits on data cells.